



EE 304: Communication Systems

Course Code	EE 304
Course Name	Communication Systems
L-T-P-C	3-0-2-4
Prerequisites	IC260, IC252
Intended for	BTech EE (core), BTech CSE (elective)
One-time approval	4th Senate (May 2013)

Course Contents

- **Signals in communication systems and their representation:** Time and frequency domain representations of signals, vector representation of signals, Constellation diagram, Baseband and bandpass signals, Amplitude and angle modulated signals (AM, FM, PM, ASK, FSK, PSK, PAM etc.), random processes and their analysis, Energy and Power spectral densities. (15 Lectures)
- **Systems in communication system and their analysis:** Modulators and demodulators (coherent and noncoherent; envelope detectors, PLLs, balanced discriminators etc.), Pre-emphasis and de-emphasis, Sampler, Quantizer, Equalizers, Encoders (PCM), Line coders (On-off, Polar, Bipolar, NRZ, RZ etc.), Pulse shapers, Bandlimited and distortion-less channels, LTI systems and random processes, Matched filter, Correlation receiver. (15 Lectures)
- **Performance analysis of analog and digital communication systems:** Signal-to-Noise Ratio (SNR) calculation for different analog communication systems, SNR analysis for PCM, Bit Error Rate (BER) calculations for different digital communication systems. (8 Lectures)
- **Case studies:** A brief overview of modem communication/broadcast technologies, e.g. Digital Radio Mondiale (DRM), Wifi, Cellular communication etc. (4 Lectures)

Text Books

1. B. P. Lathi and Z. Ding, **Modem Digital and Analog Communication Systems**, 4th Edition, Oxford Univ. Press, January 2009.
2. J. G. Proakis and M. Salehi, **Fundamentals of Communication Systems**, Prentice Hall, December 2004

References

1. S. Haykin and M. Moher, **An Introduction to Analog and Digital Communications**, 2nd Edition, Wiley, January 2006.
2. R. G. Gallager, **Principles of Digital Communication**, Cambridge University Press, March 2008.
3. A. Lapidoth, **A Foundation in Digital Communication**, Cambridge University Press, August 2009.